1010 1010 1010
1011 1011 1011
0110 0110 0101
xor 1010 1010 1010
1100 1100 1111

c s 12 s 15 bbb

swap byte (w/ carry bitmask)

a = 1010
+ 1010
0101 0100

a 0101 8-byte

guess: carries only bot but a word

```
mov     rdx, 0xaaaaaaaaaaaaaaaa
add     rax, rdx
xor     rax, rdx
```

⇒ 0xaa...aaa8

stable/same

ie $b_{63} \cdots b_2 \; b_1 b_0$

swaps each bit &
applies carry to b_{i+1}
if $b_i = 1$

ie $b_k = \bigvee_{j=1}^{k} \left(b_{2j-1} \wedge \bigwedge_{m=j+1} b_{2m-2} \right)$

$1 \to 1$

$2 \to 1$

0x39

$a = 1010$

```
mov   rdx, 0xaaaa....aaa
add   rax, rdx
xor   rax, rdx
```

$\sim 1010101010\cdots1010$

$rax = (rax_0 + 1010\cdots1010) \oplus 1010\cdots1010$

ie

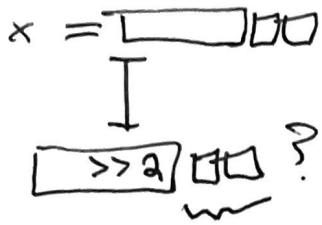
$$\begin{array}{r} \boxed{} \\ +\ 1010\cdots\quad 1010 \\ \hline \oplus\ 1010\cdots\quad 1010 \end{array}$$

$$\begin{array}{r} \cdots(1+b_3)\,b_2\,(1+b_1)\,b_0 \\ \oplus\ 1\quad 0\quad 1\quad 0 \\ \hline b_0 \end{array}$$

b_1

b_2

b_0, b_0 ! STABLE.

$x =$ 

stays same ??

NO

propagate swap ?

swap

$$\begin{array}{r} (1+b_1) \\ \oplus\ 1 \\ \hline \end{array}$$

ie $b_1 \to b_1$

$b_1 = 0$: $1 \oplus 1 = 0$ ie $b_1(0) = 0$

$b_1 = 1$: $0 \oplus 1 = 1$ ie $b_1(1) = 1$.

$b_2 \oplus 0$

$b_2 = 0$: $1 \oplus 0 = 1$

$b_2 = 1$: $0 \oplus 0 = 0$

else: b_2

special carry logic

$$b_k = \bigvee_{j=1}^{k} \left(b_{2j-1} \wedge \bigwedge_{m=j+1} b_{2m-2} \right)$$

ie

$\underbrace{b_{63} \cdots b_3\, b_2}_{\text{swap-carry bitmask}}\, \boxed{b_1, b_0}$